25AIMLPCC304IAE-II

n 3 OCT 2026

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Artificial Intelligence and Machine Learning - SEMESTER VII

25AIMLPCC304 - Database Management System

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

St. John College of Engineering & Management SJCEM PALGHAR

Q1	Attempt any 5 out of 6 Questions	Maximum marks(5)	Course Outcome	RBT Level
	a)Consider Student(Student_ID, Phone, Hobby). If a student can have several independent phones and hobbies, what type of dependency may exist?	1	CO4	L2
	b)A transaction reads all employees whose salary is greater than ₹50,000. Another transaction inserts an employee satisfying this condition. What problem can occur?	1	CO5	L2
	c)Why is Db2 suitable for both transactional and analytical workloads?	1	CO6	L2
	d)How does 3NF differ from 2NF?	1	CO4	L2
	e)A bank transfer debits ₹5,000 from one account but fails before crediting the other account. Which ACID property ensures that the transaction is rolled back?	1	CO5	L2
	f)If an application requires strong enterprise support and scalability, which Db2 characteristics should be evaluated?	1	CO6	L2
Q2	a)Define 1NF and explain the conditions that a relation must satisfy to be in First Normal Form.	5	CO4	L2
	OR			

	b)Explain the difference between lossless-join decomposition and dependency-preserving decomposition. Why are both important when decomposing a relation?	5	CO4	L4
Q3	a)Analyze the statement: "Equivalent relational algebra expressions can have different execution costs." Explain with a suitable example.	5	CO5	L4
	OR			
	b)Explain the timestamp-based concurrency control protocol. Describe the role of Read Timestamp (RTS) and Write Timestamp (WTS).	5	CO5	L3
Q4	a)Explain the important hardware and software requirements that should be considered before installing IBM Db2.	5	CO6	L3
	OR			
	b)Explain how constraints maintain data integrity in Db2. Analyze the consequences of not using appropriate constraints in a database.	5	CO6	L4